

Finite-Depth Entrance versus Stable-Manifold Renormalization Group Control

A structural companion to a macroscopic 4D Yang–Mills mass-gap construction

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February 26, 2026 (R7)

Abstract

In constructive and rigorous approaches to quantum field theory, renormalization group (RG) control is often organized around a dynamical template: an RG map acting on a space of effective actions, a fixed point or critical surface, and a stable manifold whose global geometry encodes the tuned ultraviolet trajectory. This note isolates a different proof primitive that is sufficient for a macroscopic mass-gap conclusion. The primitive is an *entrance condition*: after a finite, explicitly bounded number of blocking steps, the exact blocked specification enters a uniform Dobrushin contraction basin. From that point on, uniqueness and exponential clustering follow by standard contraction technology, and the Osterwalder–Schrader (OS) semigroup converts Euclidean exponential decay into a spectral gap for the reconstructed Hamiltonian.

We formulate an abstract *entrance-to-gap* implication for a local net of bounded gauge-invariant observables at a fixed macroscopic scale $\ell > 0$. We then explain how a concrete finite-depth entrance mechanism based on compact-group heat-kernel mixing and an explicit L^2 transfer budget replaces stable-manifold arguments in a recent Route 1 construction for four-dimensional lattice Yang–Mills. The discussion is written in standard mathematical-physics language and is intended to clarify which RG ingredients are logically necessary for the macroscopic mass-gap statement.

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1 Purpose and scope

1.1 What this note is (and is not)

This note is a structural companion to a technical construction (“Route 1”) establishing a strictly positive spectral gap for a continuum Osterwalder–Schrader (OS) reconstruction on a bounded macroscopic observable algebra at fixed positive scale $\ell > 0$. The technical work proves a *finite-depth entrance estimate*: after an explicit blocking factor $B(\beta)$, the exact blocked lattice specification lies inside a uniform Dobrushin contraction basin. It then derives exponential clustering and extracts a Hamiltonian spectral gap from the OS semigroup.

The purpose here is not to change or add to that mathematics. Instead, we isolate the minimal RG control needed for the macroscopic gap statement and compare it with the traditional stable-manifold RG template.

1.2 What is made explicit in this note

The note is a companion and does not introduce new model-dependent estimates. Its contribution is to make the logical implication chain and its hypotheses fully explicit in a way that can be cited independently:

- (C1) an abstract “entrance-to-gap” theorem stated on a bounded macroscopic OS-positive net, with a complete proof using OS reconstruction and a semigroup spectral argument;
- (C2) an explicit lemma chain transferring one-site conditional χ^2 control to Dobrushin contractivity (Appendix A);
- (C3) a dictionary table recording how the Route 1 modules map to the abstractions used here, with each entry marked by logical status.

1.3 Macroscopic observable net

Fix a physical scale $\ell > 0$ and let $\mathcal{R}$ denote the family of bounded open rectangles in $\mathbb{R}^4$ (any convenient base of bounded regions suffices). A *bounded macroscopic observable net at scale ℓ* is an assignment

$$\Lambda \in \mathcal{R} \mapsto \mathcal{O}_\ell(\Lambda)$$

of unital $*$ -algebras of bounded, gauge-invariant observables such that:

- (N1) *Isotony*. If $\Lambda_1 \subseteq \Lambda_2$ then $\mathcal{O}_\ell(\Lambda_1) \subseteq \mathcal{O}_\ell(\Lambda_2)$.

- (N2) *Euclidean translation covariance.* There is an action $(\tau_x)_{x \in \mathbb{R}^4}$ by *-automorphisms such that $\tau_x \mathcal{O}_\ell(\Lambda) = \mathcal{O}_\ell(\Lambda + x)$.
- (N3) *Time reflection covariance.* With $\theta(t, \mathbf{x}) := (-t, \mathbf{x})$, there is a time-reflection anti-automorphism Θ such that $\Theta \mathcal{O}_\ell(\Lambda) = \mathcal{O}_\ell(\theta\Lambda)$.

We set

$$\mathcal{A} := \bigcup_{\Lambda \in \mathcal{R}} \mathcal{O}_\ell(\Lambda), \quad \mathcal{A}_+ := \bigcup_{\Lambda \subset \{t > 0\}} \mathcal{O}_\ell(\Lambda).$$

The macroscopic mass-gap statement below concerns the spectral gap of the OS-reconstructed Hamiltonian obtained from an OS-positive functional on $\mathcal{A}$, acting on the Hilbert space generated by $\mathcal{A}_+$. No claim is made here about sharp pointlike fields or $t \rightarrow 0$ limits.

Remark 1 (Lattice realizations). *In the lattice setting motivating Route 1, $\mathcal{O}_\ell(\Lambda)$ may be taken to be generated by bounded gauge-invariant cylinder observables whose supports lie in a discretization of Λ and have physical diameter $O(\ell)$ under the chosen identification. The present note treats the net abstractly and uses only (N1)–(N3) together with OS positivity on $\mathcal{A}$.*

2 Two RG control templates

2.1 The stable-manifold template

In the conventional constructive RG template one introduces:

- (SM1) a (typically infinite-dimensional) space $\mathfrak{S}$ of effective actions or specifications;
- (SM2) an RG map $\mathcal{R} : \mathfrak{S} \rightarrow \mathfrak{S}$ encoding a blocking step;
- (SM3) a distinguished object $S_* \in \mathfrak{S}$ (fixed point, critical surface, or scaling solution);
- (SM4) a decomposition of perturbations into relevant/irrelevant directions and a proof that irrelevant directions contract under iteration.

This template (in various guises) is classical; see for example Wilson and Kogut [6] for the physics formulation, Brydges [7] for a rigorous treatment including the stable-manifold viewpoint, and Bałaban [8, 9] for a renormalization group approach to lattice gauge field theories. One then tunes the ultraviolet initial condition so that the RG trajectory stays on (or enters) a stable manifold of S_* . The global trajectory organizes control across arbitrarily many scales.

Remark 2 (What stable-manifold control typically delivers). *Beyond existence of a limit theory, stable-manifold control can deliver quantitative information about the scaling of couplings, universality classes, and regularity upgrades (e.g. sharp-field reconstruction) when combined with additional estimates. The point of the present note is that the macroscopic gap statement on $\mathcal{O}_\ell$ requires strictly less dynamical information.*

2.2 Finite-depth entrance as a different primitive

The finite-depth entrance primitive replaces global trajectory control by a one-shot estimate. Fix a contraction region $\mathfrak{B} \subset \mathfrak{S}$ (for example, the set of specifications with Dobrushin constant less than 1). The entrance condition is:

There exists a finite depth k (possibly depending on the microscopic coupling) such that $\mathcal{R}^k(S)$ lies in $\mathfrak{B}$.

Once inside $\mathfrak{B}$, standard contraction consequences (uniqueness, exponential mixing) hold without further dynamical structure. The central difference is logical: a stable-manifold proof builds a global object (a trajectory staying in a good region forever), whereas an entrance proof produces a single inequality placing the blocked object in a good region after finitely many steps.

Remark 3 (Entrance is a one-shot substitute for global trajectory control). *Any RG argument that constructs a trajectory $\{\mathcal{R}^n(S)\}_{n \geq 0}$ which enters a contraction region $\mathfrak{B}$ at some finite time and remains in $\mathfrak{B}$ thereafter yields a finite-depth entrance estimate as a corollary. Conversely, a finite-depth entrance estimate is a single inequality and does not, by itself, produce global invariant-manifold structure (e.g. parameterizations of tuned trajectories or universality information). The point here is that the macroscopic spectral-gap extraction on $\mathcal{O}_\ell$ requires only the one-shot entrance data.*

3 Contraction technology and the entrance-to-gap implication

All statements in this section are standard in Gibbs specification theory and OS reconstruction. We record them in a form convenient for the comparison.

3.1 Dobrushin contraction

Let Γ be a specification on a lattice $\mathbb{Z}^d$ (or on a block lattice) with single-site conditional kernels $\gamma_x(\cdot \mid \eta)$. Let C_{xy} denote the Dobrushin influence matrix and set

$$\delta(\Gamma) := \sup_x \sum_{y \neq x} C_{xy}.$$

We say Γ is *Dobrushin contractive* if $\delta(\Gamma) < 1$.

Theorem 1 (Dobrushin uniqueness and exponential clustering). *If a specification Γ is Dobrushin contractive, then: $[1, 2]$*

(D1) *there exists a unique infinite-volume Gibbs state consistent with Γ ;*

(D2) *truncated correlations of bounded local observables decay exponentially in distance.*

3.2 A perturbation bound for contraction

One way to obtain contractivity of an *exact* specification is to compare it to a *reference* contractive specification. The details vary with the model, but a typical structural statement is:

Proposition 1 (Transfer of Dobrushin contractivity under a uniform χ^2 gate). *Let Γ^{ref} and Γ^{ex} be specifications with single-site conditional kernels $\gamma_x^{\text{ref}}(\cdot \mid \eta)$ and $\gamma_x^{\text{ex}}(\cdot \mid \eta)$ on the same configuration space. Assume there exists a family of finite neighbourhoods $N(x)$ with*

$$\Delta := \sup_x |N(x) \setminus \{x\}| < \infty$$

such that, for both specifications, $\gamma_x(\cdot \mid \eta)$ depends on η only through $\eta|_{N(x) \setminus \{x\}}$. Assume $\delta(\Gamma^{\text{ref}}) \leq q_ < 1$.*

(a) *If*

$$\sup_x \sup_\eta \|\cdot\|_{\text{TV}}(\gamma_x^{\text{ex}}(\cdot \mid \eta), \gamma_x^{\text{ref}}(\cdot \mid \eta)) \leq \varepsilon,$$

then

$$\delta(\Gamma^{\text{ex}}) \leq q_* + 2\Delta \varepsilon.$$

(b) If, moreover, $\gamma_x^{\text{ex}}(\cdot | \eta) \ll \gamma_x^{\text{ref}}(\cdot | \eta)$ and

$$\sup_x \sup_{\eta} D_{\chi^2}(\gamma_x^{\text{ex}}(\cdot | \eta) \| \gamma_x^{\text{ref}}(\cdot | \eta)) \leq \alpha^2,$$

then

$$\delta(\Gamma^{\text{ex}}) \leq q_* + \Delta \alpha.$$

In particular, Γ^{ex} is Dobrushin contractive whenever $\Delta \alpha < 1 - q_*$.

Proof. See Appendix A. □

Remark 4 (How this transfer is used in Route 1). *The Route 1 construction establishes a bound of type (b) at an explicit blocking depth by a multiplicative second-moment estimate along the blocking filtration (a uniform conditional χ^2 bound for one-site kernels). The contractivity of the exact blocked specification then follows immediately from Proposition 1.*

Remark 5 (Neighbourhood locality at a block scale). *For block-spin/block-gauge transformations of fixed finite geometry, the blocked specification at any finite blocking depth is local with respect to a neighbourhood structure determined by block adjacency. In particular, the dependence degree Δ in Proposition 1 is a purely geometric constant (depending only on dimension and the blocking geometry), and is uniform in volume. This is the setting in which the χ^2 gate is used in Route 1.*

3.3 Finite-depth entrance and clustering

Consider a family of microscopic lattice theories indexed by a coupling parameter (e.g. inverse coupling β) and a blocking map $\mathcal{R}$ acting on the corresponding specifications.

Definition 1 (Finite-depth entrance). *Fix a contraction basin parameter $q \in (0, 1)$. A family of specifications $\{\Gamma_\beta\}_{\beta \geq \beta_0}$ has a finite-depth entrance into Dobrushin contraction if there exists a function $k(\beta) \in \mathbb{N}$ such that*

$$\delta(\mathcal{R}^{k(\beta)}(\Gamma_\beta)) \leq q < 1 \quad \text{for all } \beta \geq \beta_0.$$

Corollary 1 (Entrance-to-clustering). *If $\{\Gamma_\beta\}$ has finite-depth entrance into Dobrushin contraction, then each blocked theory $\mathcal{R}^{k(\beta)}(\Gamma_\beta)$ has a unique infinite-volume Gibbs state and exponential clustering at the block scale.*

3.4 From macroscopic Euclidean decay to a Hamiltonian spectral gap

We record an OS semigroup setup and a spectral criterion in a form convenient for a bounded macroscopic observable net.

3.4.1 OS setup on a bounded net

Let $\mathcal{A}$ be a unital $*$ -algebra of bounded Euclidean observables equipped with:

- (OS1) a time-reflection anti-automorphism $\Theta : \mathcal{A} \rightarrow \mathcal{A}$;
- (OS2) a one-parameter group of Euclidean time translations $(\tau_t)_{t \in \mathbb{R}}$ acting by $*$ -automorphisms of $\mathcal{A}$; in the net setting of Section 1.3 we take $\tau_t := \tau_{(t, \mathbf{0})}$ as the restriction of the translation action $(\tau_x)_{x \in \mathbb{R}^4}$, so that $\tau_t \mathcal{A}_+ \subseteq \mathcal{A}_+$ for all $t \geq 0$;
- (OS3) a distinguished positive-time subalgebra $\mathcal{A}_+ \subset \mathcal{A}$.

Remark 6 (Compatibility with the net structure). *In the bounded-net setting of Section 1.3, the OS data are induced by the net covariance: Θ is the time-reflection from (N3), $\tau_t := \tau_{(t, \mathbf{0})}$ is the restriction of the translation action from (N2) to the time axis, and $\mathcal{A}_+$ is the union of algebras localized in the positive-time half-space. In particular, $\tau_t \mathcal{A}_+ \subseteq \mathcal{A}_+$ for all $t \geq 0$.*

Let $\mathcal{S}$ be a linear functional on $\mathcal{A}$. We say $\mathcal{S}$ is *OS-positive* if

$$\mathcal{S}(\Theta F F) \geq 0 \quad \forall F \in \mathcal{A}_+. \quad (1)$$

Assume also translation invariance $\mathcal{S}(\tau_t F) = \mathcal{S}(F)$. Under (1), the OS inner product on $\mathcal{A}_+$ is

$$\langle F, G \rangle_{\text{OS}} := \mathcal{S}(\Theta F G). \quad (2)$$

Quotienting by null vectors and completing yields a Hilbert space $\mathcal{H}$ and a cyclic vector Ω given by the class of $\mathbf{1}$. For $t \geq 0$ define an operator T_t on the dense subspace $\mathcal{A}_+/\mathcal{N}$ by

$$T_t[F] := [\tau_t F].$$

Translation invariance gives $\|T_t[F]\|^2 = \mathcal{S}(\Theta \tau_t F \tau_t F) = \mathcal{S}(\Theta F F) = \|F\|^2$, so each T_t extends to a contraction on $\mathcal{H}$ and $\{T_t\}_{t \geq 0}$ is a strongly continuous semigroup. By the OS reconstruction theorem there exists a self-adjoint generator $H \geq 0$ such that $T_t = e^{-tH}$. [3,4]

Lemma 1 (OS semigroup matrix elements). *For $F, G \in \mathcal{A}_+$ and $t \geq 0$,*

$$\langle [F], e^{-tH}[G] \rangle_{\mathcal{H}} = \mathcal{S}(\Theta F \tau_t G).$$

Proof. By definition of the OS inner product (2) and the semigroup $T_t[F] = [\tau_t F]$ constructed above,

$$\langle [F], e^{-tH}[G] \rangle_{\mathcal{H}} = \langle [F], T_t[G] \rangle_{\mathcal{H}} = \mathcal{S}(\Theta F \tau_t G).$$

□

Lemma 2 (A semigroup spectral-gap criterion on a dense subspace). *Let $H \geq 0$ be self-adjoint on a Hilbert space $\mathcal{H}$ with $H\Omega = 0$. Let $\mathcal{D} \subset \mathcal{H}$ be a dense linear subspace containing Ω . Assume there exist $m > 0$ such that for every $\psi \in \mathcal{D}$ with $\langle \Omega, \psi \rangle = 0$ there exists $C_\psi < \infty$ satisfying*

$$\langle \psi, e^{-tH} \psi \rangle \leq C_\psi e^{-mt} \quad \forall t \geq 0.$$

Then

$$\text{Spec}(H) \subset \{0\} \cup [m, \infty).$$

Proof. Fix $\varepsilon \in (0, m)$ and let $P := \mathbf{1}_{[0, m-\varepsilon]}(H)$ be the spectral projection. Suppose $\text{Ran}(P)$ contains a vector orthogonal to Ω . Since $\mathcal{D}$ is dense, choose $\psi \in \mathcal{D}$ such that $P\psi \notin \text{span}\{\Omega\}$. Replace ψ by $\psi - \langle \Omega, \psi \rangle \Omega$ so that $\psi \perp \Omega$ while still $P\psi \neq 0$. Then

$$\langle \psi, e^{-tH} \psi \rangle \geq \langle P\psi, e^{-tH} P\psi \rangle \geq e^{-(m-\varepsilon)t} \|P\psi\|^2.$$

On the other hand, by hypothesis there is $C_\psi < \infty$ with $\langle \psi, e^{-tH} \psi \rangle \leq C_\psi e^{-mt}$. For large t this is impossible because $e^{\varepsilon t} \|P\psi\|^2 / C_\psi \rightarrow \infty$. Hence $\text{Ran}(P) \subset \text{span}\{\Omega\}$ for all $\varepsilon \in (0, m)$, which implies $\text{Spec}(H) \subset \{0\} \cup [m, \infty)$. □

3.5 Entrance-to-gap on a macroscopic OS-positive net

Assumption 1 (OS-positive macroscopic limit). *Fix $\ell > 0$. Assume that along a chosen ultraviolet identification (a sequence of lattice spacings and couplings realizing the scale ℓ), the joint laws of bounded local observables in $\mathcal{O}_\ell$ converge to a translation-invariant OS-positive functional $\mathcal{S}$ on a bounded net $\mathcal{A} \supset \mathcal{O}_\ell$ equipped with OS data $(\Theta, \tau_t, \mathcal{A}_+)$ as in (1)–(2).*

Theorem 2 (Entrance-to-gap principle (macroscopic)). *Assume: (i) finite-depth entrance into a uniform Dobrushin contraction basin; and (ii) Assumption 1. Then in the OS reconstruction $(\mathcal{H}, \Omega, H)$ associated with the limiting Schwinger functional on $\mathcal{O}_\ell$ there exists $m > 0$ such that*

$$\text{Spec}(H) \subset \{0\} \cup [m, \infty).$$

Proof. Let Γ_β denote the (blocked) specification at the entrance scale. By Corollary 1 and Theorem 1, for each sufficiently large β the corresponding infinite-volume Gibbs state has exponential clustering for bounded local observables. In particular, for bounded observables F, G supported in disjoint regions separated by Euclidean time distance t one has

$$|\langle FG \rangle_c| \leq C_{F,G} e^{-mt}, \quad \langle FG \rangle_c := \langle FG \rangle - \langle F \rangle \langle G \rangle, \quad (3)$$

for some $m > 0$ (uniform at the entrance scale) and constants $C_{F,G} < \infty$.

Passing to the macroscopic limit along the ultraviolet identification in Assumption 1 yields the same form of exponential time-clustering for the limiting functional $\mathcal{S}$ on $\mathcal{A}$. Let $F \in \mathcal{A}_+$ and write $\tilde{F} := F - \mathcal{S}(F)\mathbf{1}$. Then $\langle \Omega, [\tilde{F}] \rangle = 0$ and, by Lemma 1,

$$\langle [\tilde{F}], e^{-tH} [\tilde{F}] \rangle = \mathcal{S}(\Theta \tilde{F} \tau_t \tilde{F}) = \mathcal{S}(\Theta F \tau_t F) - \mathcal{S}(\Theta F) \mathcal{S}(F).$$

The right-hand side is a truncated correlation across a Euclidean time separation t , hence by (3)

$$\langle [\tilde{F}], e^{-tH} [\tilde{F}] \rangle \leq C_F e^{-mt} \quad \forall t \geq 0. \quad (4)$$

Let $\mathcal{D} := \text{span}\{[\tilde{F}] : F \in \mathcal{A}_+\}$. By OS reconstruction, $\mathcal{D}$ is dense in $\mathcal{H}$. Applying Lemma 2 with $\mathcal{D}$ and the bound (4) gives $\text{Spec}(H) \subset \{0\} \cup [m, \infty)$. $\square$

Remark 7 (What is and is not asserted). *Theorem 2 is an implication for the macroscopic net $\mathcal{O}_\ell$. It does not assert existence of sharp pointlike fields. It also does not assert that a unique macroscopic scaling limit exists without the explicit ultraviolet identification used to formulate Assumption 1. It asserts that once a macroscopic OS-positive limit exists and macroscopic exponential clustering holds, a Hamiltonian spectral gap follows on that reconstructed theory.*

4 Finite-depth entrance mechanisms without stable manifolds

The entrance condition can be obtained by different methods. Here we isolate a modular pattern that arises naturally in compact-group lattice gauge theory.

4.1 Mixing witness at the group level

A convenient reference object is the heat-kernel density on a compact Lie group G . Its convolution powers exhibit explicit χ^2 -mixing at a finite semigroup time. The key feature is that the mixing depth can be bounded directly in terms of the Casimir spectrum. This produces an explicit *mixing witness* for the reference specification.

4.2 An L^2 transfer budget

To compare the exact blocked specification to the reference one, one controls the Radon–Nikodym defect along a finite blocking schedule in an L^2 sense. A typical step has the form

$$\frac{d\Gamma_{n+1}^{\text{ex}}}{d\Gamma_{n+1}^{\text{ref}}} = \frac{d\Gamma_n^{\text{ex}}}{d\Gamma_n^{\text{ref}}} r_{n+1}, \quad \mathbb{E}_{\Gamma_{n+1}^{\text{ref}}} [r_{n+1}] = 1,$$

with a per-step bound $\|r_{n+1} - 1\|_2 \leq \eta_{n+1}$. A finite-depth transfer budget asserts $\sum_{j \leq k} \eta_j$ (or $\sum \eta_j^2$) is small at the entrance depth.

4.3 From L^2 control to a χ^2 gate

The crucial step is to pass from the transfer budget to a bound on the χ^2 divergence between exact and reference conditionals. This is done by a multiplicative second-moment estimate along a blocking filtration. The resulting χ^2 gate implies proximity in total variation, and therefore transfers contraction from the reference specification to the exact one via a stability estimate such as Proposition 1.

Remark 8 (Modularity). *This “mixing witness + transfer budget + χ^2 gate” pattern is logically independent of stable-manifold RG. It produces a one-shot entrance inequality at a computable scale. The subsequent consequences (uniqueness, clustering, OS spectral gap) follow by standard contraction and semigroup arguments.*

5 Relation to the Route 1 Yang–Mills construction

In the Route 1 technical manuscript, the above entrance pattern is implemented for four-dimensional lattice Yang–Mills theory with compact connected simple gauge group and Wilson action. The reference specification is built from a heat-kernel benchmark at parameter $t \sim 1/\beta$, whose convolution powers yield explicit χ^2 mixing after a semigroup time s_δ and hence an explicit blocking scale $B(\beta)$. A per-step L^2 transfer estimate controls the defect between the exact blocked specification and the reference one. A multiplicative second-moment lemma yields a χ^2 gate, and therefore places the exact blocked specification in a uniform Dobrushin contraction basin.

The macroscopic mass-gap statement then follows by combining:

- (R1) finite-depth entrance $\Rightarrow$ exponential clustering at the block scale (Theorem 1);
- (R2) an explicit ultraviolet identification at fixed macroscopic scale ℓ to express decay in physical units;
- (R3) OS positivity and semigroup spectral extraction (Theorem 2).

5.1 Dictionary between Route 1 modules and the abstractions used here

Table 1 records how the named modules and estimates in the Route 1 manuscript correspond to the abstract hypotheses and lemmas used in this note. The table is organizational: it introduces no additional assumptions beyond those already stated.

Remark 9 (Status of the reference contractivity input). *In the Route 1 manuscript, contractivity of the chosen reference specification is stated explicitly as an assumption (Ass. R1). One sufficient checkable criterion is uniform near-Haar control of the one-site kernels: if $\sup_x \sup_\eta D_{\chi^2}(\gamma_x^{\text{ref}}(\cdot | \eta) \| dg) \leq \varepsilon_{\text{Haar}}$, then the elementary estimate in Appendix A (Lemma 4) yields $\delta(\Gamma^{\text{ref}}) \leq \Delta \sqrt{\varepsilon_{\text{Haar}}}$. In particular, when the reference kernels are implemented via a compact-group heat-kernel benchmark at a mixing time for which $D_{\chi^2}(K_t \| dg)$ is small, this criterion gives an explicit contraction constant $q_* < 1$.*

The conceptual point is that no stable manifold is constructed and no global RG trajectory analysis is required. The only RG “time” used is the finite number of steps needed to enter the contraction basin.

6 Discussion: what stable-manifold data are (not) needed for a macroscopic gap

A stable-manifold construction typically provides more dynamical information than is needed for the macroscopic mass-gap statement: parameterization of trajectories, control of relevant

Route 1 module / estimate	Abstraction in this note	Logical role in the entrance-to-gap chain	Status
Heat-kernel benchmark on G with explicit χ^2 mixing depth; chosen reference specification Γ^{ref}	Reference contractivity $\delta(\Gamma^{\text{ref}}) \leq q_* < 1$ in Proposition 1	Provides a uniform contraction basin to which the exact blocked specification will be compared.	Assumed in Route 1 (Ass. R1)
Per-step L^2 defect bounds along a finite blocking schedule (transfer budget)	Uniform per-site conditional χ^2 bound (a “ χ^2 gate”) in Proposition 1 (b) and Appendix A	Ensures the exact one-site kernels are close to the reference kernels uniformly in boundary conditions, enabling transfer of Dobrushin contractivity.	Conditional in Route 1 (Ass. 3)
Multiplicative second-moment bound along the blocking filtration (accumulation control)	Mechanism producing the χ^2 gate (abstracted here as the hypothesis of Proposition 1 (b))	Converts a finite-depth transfer budget into uniform conditional divergence control at the entrance depth.	Proved in Route 1
Entrance theorem: explicit blocking depth $k(\beta)$ (or $B(\beta)$) after which the exact blocked specification is contractive	Finite-depth entrance (Definition 1) and entrance-to-clustering (Corollary 1)	Upgrades local kernel control to global uniqueness and exponential clustering for macroscopic observables.	Proved in Route 1 (under assumptions)
OS-positive macroscopic scaling limit on the bounded net at scale ℓ	Assumption 1	Provides OS reconstruction and a positivity-preserving semigroup e^{-tH} on the reconstructed Hilbert space.	External input
Euclidean exponential decay for macroscopic observables (physical units)	Semigroup gap criterion (Lemma 2) and Theorem 2	Converts Euclidean clustering into a strictly positive spectral gap for H on the reconstructed theory.	Standard consequence
UV identification inputs (mapping $\beta \mapsto a(\beta)$ and non-collapse of the physical decay rate)	Contained in the formulation of Assumption 1 and in the passage “block scale $\rightarrow$ physical units” in Theorem 2	Allows expressing the contraction-derived decay rate as a nonzero mass parameter in physical units at fixed ℓ .	External input

Table 1: Dictionary between Route 1 modules and the abstract assumptions/lemmas used in this note.

couplings at all scales, and a dynamical organizing center. For the macroscopic gap on $\mathcal{O}_\ell$, the essential ingredients reduce to:

- (E1) a local net of bounded observables at fixed scale ℓ ;
- (E2) OS positivity of the limiting Schwinger functional on that net;
- (E3) exponential clustering (uniform in volume) for those observables.

Finite-depth entrance provides (E3) once the net is identified with a blocked lattice scale. The ultraviolet identification required to formulate (E1)–(E2) is explicitly separated in Route 1 and is orthogonal to the entrance mechanism.

A A χ^2 -to-Dobrushin transfer lemma chain

This appendix records a standard lemma chain used to transfer Dobrushin contractivity from a reference specification to an exact specification when one has conditional χ^2 control of the one-site kernels. The statements are elementary, but writing them explicitly makes the contractivity-transfer step in the finite-depth entrance mechanism fully transparent.

A.1 Total variation and χ^2 divergence

Let μ and ν be probability measures on a measurable space $(\Omega, \mathcal{F})$. Their total variation distance is

$$\|\cdot\|_{\text{TV}}(\mu, \nu) := \sup_{A \in \mathcal{F}} |\mu(A) - \nu(A)| = \frac{1}{2} \int_{\Omega} \left| \frac{d\mu}{d\lambda} - \frac{d\nu}{d\lambda} \right| d\lambda,$$

for any dominating measure λ . If $\mu \ll \nu$, the χ^2 divergence is

$$D_{\chi^2}(\mu \| \nu) := \int_{\Omega} \left(\frac{d\mu}{d\nu} - 1 \right)^2 d\nu.$$

Lemma 3 (χ^2 controls total variation). *If $\mu \ll \nu$ then*

$$\|\cdot\|_{\text{TV}}(\mu, \nu) \leq \frac{1}{2} \sqrt{D_{\chi^2}(\mu \| \nu)}.$$

Proof. Let $r := d\mu/d\nu$. Then $\|\cdot\|_{\text{TV}}(\mu, \nu) = \frac{1}{2} \int |r - 1| d\nu$. By Cauchy–Schwarz, $\int |r - 1| d\nu \leq (\int (r - 1)^2 d\nu)^{1/2}$, giving the claim. $\square$

A.2 Locality and influence matrices

Let Γ be a specification with one-site kernels $\gamma_x(\cdot | \eta)$. For $x \neq y$ define the Dobrushin influence coefficient

$$C_{xy}(\Gamma) := \sup \left\{ \|\cdot\|_{\text{TV}}(\gamma_x(\cdot | \eta), \gamma_x(\cdot | \eta')) : \eta_z = \eta'_z \ \forall z \neq y \right\},$$

and the Dobrushin constant $\delta(\Gamma) := \sup_x \sum_{y \neq x} C_{xy}(\Gamma)$.

We will use a standard locality hypothesis ensuring that only finitely many sites can influence a given conditional.

Definition 2 (Uniform neighbourhood locality). *A specification Γ is said to be local with neighbourhoods $N(x)$ if for each site x there is a finite set $N(x) \ni x$ such that whenever two boundary conditions η, η' satisfy $\eta|_{N(x) \setminus \{x\}} = \eta'|_{N(x) \setminus \{x\}}$, one has $\gamma_x(\cdot | \eta) = \gamma_x(\cdot | \eta')$. We call*

$$\Delta := \sup_x |N(x) \setminus \{x\}|$$

the dependence degree.

Lemma 4 (Near-Haar one-site kernels imply Dobrushin contraction). *Let Γ be local with neighbourhoods $N(x)$ and dependence degree $\Delta < \infty$ (Definition 2). Assume there exists $\varepsilon_{\text{Haar}} \geq 0$ such that for all sites x and boundary conditions η ,*

$$D_{\chi^2}(\gamma_x(\cdot | \eta) \| dg) \leq \varepsilon_{\text{Haar}},$$

where dg denotes Haar measure on the one-site space. Then

$$\delta(\Gamma) \leq \Delta \sqrt{\varepsilon_{\text{Haar}}}.$$

In particular, Γ is Dobrushin contractive whenever $\Delta \sqrt{\varepsilon_{\text{Haar}}} < 1$.

Proof. By Lemma 3, $\|\cdot\|_{\text{TV}}(\gamma_x(\cdot | \eta), dg) \leq \frac{1}{2} \sqrt{\varepsilon_{\text{Haar}}}$ uniformly in x, η . If η, η' differ at a single site, then by the triangle inequality,

$$\|\cdot\|_{\text{TV}}(\gamma_x(\cdot | \eta), \gamma_x(\cdot | \eta')) \leq \|\cdot\|_{\text{TV}}(\gamma_x(\cdot | \eta), dg) + \|\cdot\|_{\text{TV}}(dg, \gamma_x(\cdot | \eta')) \leq \sqrt{\varepsilon_{\text{Haar}}}.$$

Thus $C_{xy}(\Gamma) \leq \sqrt{\varepsilon_{\text{Haar}}}$ for all $x \neq y$. Locality implies $C_{xy}(\Gamma) = 0$ unless $y \in N(x) \setminus \{x\}$, so summing over at most Δ sites yields $\delta(\Gamma) \leq \Delta \sqrt{\varepsilon_{\text{Haar}}}$. $\square$

Lemma 5 (Influence transfer under uniform total-variation perturbations). *Let Γ^{ref} and Γ^{ex} be specifications that are local with the same neighbourhoods $N(x)$ and dependence degree $\Delta < \infty$. Assume that for some $\varepsilon \geq 0$,*

$$\sup_x \sup_{\eta} \|\cdot\|_{\text{TV}}(\gamma_x^{\text{ex}}(\cdot | \eta), \gamma_x^{\text{ref}}(\cdot | \eta)) \leq \varepsilon.$$

Then for all $x \neq y$,

$$C_{xy}(\Gamma^{\text{ex}}) \leq C_{xy}(\Gamma^{\text{ref}}) + 2\varepsilon.$$

Proof. Fix $x \neq y$ and boundary conditions η, η' that agree off y . By the triangle inequality,

$$\begin{aligned} \|\cdot\|_{\text{TV}}(\gamma_x^{\text{ex}}(\cdot | \eta), \gamma_x^{\text{ex}}(\cdot | \eta')) &\leq \|\cdot\|_{\text{TV}}(\gamma_x^{\text{ex}}(\cdot | \eta), \gamma_x^{\text{ref}}(\cdot | \eta)) + \|\cdot\|_{\text{TV}}(\gamma_x^{\text{ref}}(\cdot | \eta), \gamma_x^{\text{ref}}(\cdot | \eta')) \\ &\quad + \|\cdot\|_{\text{TV}}(\gamma_x^{\text{ref}}(\cdot | \eta'), \gamma_x^{\text{ex}}(\cdot | \eta')) \leq \varepsilon + \|\cdot\|_{\text{TV}}(\gamma_x^{\text{ref}}(\cdot | \eta), \gamma_x^{\text{ref}}(\cdot | \eta')) + \varepsilon. \end{aligned}$$

Taking the supremum over such η, η' yields the bound. $\square$

Proposition 2 (Dobrushin constant transfer). *In the setting of Lemma 5,*

$$\delta(\Gamma^{\text{ex}}) \leq \delta(\Gamma^{\text{ref}}) + 2\Delta \varepsilon.$$

In particular, if $\delta(\Gamma^{\text{ref}}) \leq q_* < 1$ and $2\Delta \varepsilon < 1 - q_*$, then Γ^{ex} is Dobrushin contractive.

Proof. By locality, $C_{xy}(\Gamma^{\text{ex}}) = 0$ for $y \notin N(x) \setminus \{x\}$, and similarly for Γ^{ref} . Thus, for each x ,

$$\sum_{y \neq x} C_{xy}(\Gamma^{\text{ex}}) = \sum_{y \in N(x) \setminus \{x\}} C_{xy}(\Gamma^{\text{ex}}) \leq \sum_{y \in N(x) \setminus \{x\}} (C_{xy}(\Gamma^{\text{ref}}) + 2\varepsilon) \leq \sum_{y \neq x} C_{xy}(\Gamma^{\text{ref}}) + 2\Delta \varepsilon.$$

Taking $\sup_x$ yields the claim. $\square$

Corollary 2 (χ^2 gate $\Rightarrow$ contraction transfer). *Assume Γ^{ref} and Γ^{ex} are local with common neighbourhoods of dependence degree $\Delta < \infty$ and satisfy $\delta(\Gamma^{\text{ref}}) \leq q_* < 1$. If*

$$\sup_x \sup_{\eta} D_{\chi^2}(\gamma_x^{\text{ex}}(\cdot | \eta) \| \gamma_x^{\text{ref}}(\cdot | \eta)) \leq \alpha^2,$$

then $\delta(\Gamma^{\text{ex}}) \leq q_* + \Delta \alpha$. In particular, Γ^{ex} is Dobrushin contractive whenever $\Delta \alpha < 1 - q_*$.

Proof. Lemma 3 gives $\varepsilon \leq \alpha/2$, and Proposition 2 gives $\delta(\Gamma^{\text{ex}}) \leq \delta(\Gamma^{\text{ref}}) + 2\Delta(\alpha/2) = q_* + \Delta \alpha$. $\square$

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